

Technical Datasheet

KROHNE

measure the facts

H250 M40

Variable area flowmeter

- Secure, cost-effective measurement and display, even without power supply
- Modular, can be expanded right down to FOUNDATION™ Fieldbus
- Sturdy, closed design for high resistance to pressure, temperature and media
- Available in variety of process connections
- Optional module with integrated LCD display and totalizer

Product Features

Intelligent Modularity

The H250 all-metal variable area flowmeters cover the entire range of requirements of the process industry. The modular device design and flexible production structure form the basis for application and customer-based device versions.

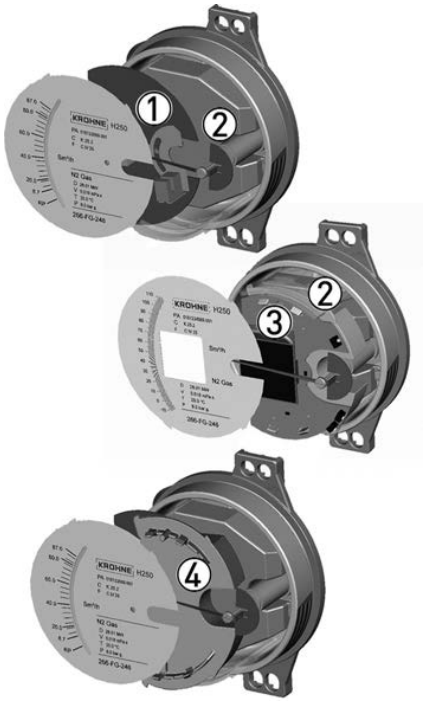


The basis of the H250 M40 is its purely mechanical design.

Additional electronic modules can be added or replaced at any time without interrupting the process.

This way, the functionality of the device adapts to any changed requirements. From analogue flow measurement without auxiliary power to digital integration into a fieldbus system.

Simply screw off the cover, insert the module until it clicks, screw the cover back on and that's it. It is just as easy to replace the measuring scale in the event of a change in application.



① 1 or 2 limit switches, type NAMUR Transistor (3-wire) or Reed

② 2-wire signal output 4...20 mA with HART

③ Additional LCD for flow rate value and/or volume counter, 2 configurable binary outputs for limit or pulse output and 1 binary input for starting/stopping or resetting the counter.

④ 2-wire fieldbus communication Foundation Fieldbus or Profibus PA

All modules are intrinsically safe (Ex-i)

Highlights

- Simple, low-cost installation: Measurement and indication without auxiliary power supply
- Universal Ex concept: Ex-i
- Modular scalability – from mechanical to fieldbus
- Robust measuring tube construction for high process temperatures and extreme operating pressures
- Choice of material: Stainless steel, PTFE
- Many connection variants: flanged, screwed, clamped
- High application safety, even with extremely low flows

Industries

Can be used in all industrial sectors, for example:

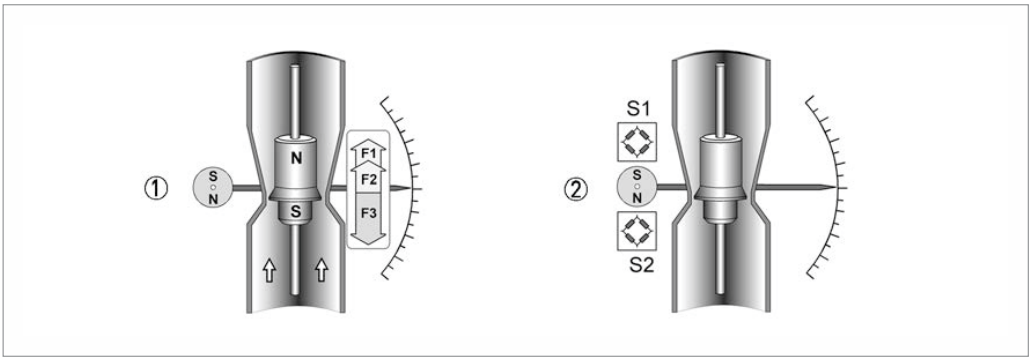
- Chemical
- Petrochemicals
- Pharma
- Machine building
- Food & Beverage
- Oil & Gas
- Iron, Steel & Metal
- Power plants
- Pulp & Paper
- Water & wastewater industries

Applications

- Nitrogen inerting to avoid explosive atmospheres
- Measurement of additives such as catalysts, tensides, anti-foaming and anti-corrosion agents
- Measurement of chlorine, sulphur and ethylene compounds
- Measurement of distilled or demineralised water
- Monitoring of lubricating and cooling agents for process pumps and rotating machines
- Monitoring of sealing systems on compressors
- Gas measurement for industrial furnaces
- Hygienic applications in the food industry and in pharmaceuticals

Functional principle

The H250 flowmeter operates in accordance with the float measuring principle. The measuring unit consists of a metal cone in which a float can move freely up and down. The flow goes from bottom to top. The float changes position so that the lifting force acting on it F1 is in equilibrium with the form drag F2 and its weight F3: F3 = F1 + F2



① Indication principle M40 magnetic coupling
② Magnetic coupling sensors

① For the indicator, the flow-dependent height of the float in the measuring unit is transmitted by means of a magnetic coupling and displayed on a scale.

② For a built-in signal converter (ESK4), the flow-dependent height of the float in the measuring unit is detected by the S1 and S2 magnetic field sensors and electronically processed.

Technical data

- The following data is provided for general applications. If you require data that is more relevant to your specific application, please contact us or your local sales office.

Measuring system

Application range	Flow measurement of liquids, gases and vapours
Function/Measuring principle	Variable area measuring principle
Measured value	
Primary measured value	Float position
Secondary measured value	Operating and standard volumetric flow

Measuring accuracy

Directive	VDI/VDE 3513, sheet 2 (q_G = 50%)
H250 / RR	1.6%
H250/PTFE	2.5%

Operating conditions

Temperature	
Max. operating temperature TS	-80..+300°C depending on the version
Pressure	
Max. operating pressure PS	depending on the version
Max. test pressure PT	depending on the version
Min. required operating pressure	2 times greater than pressure loss (see measuring ranges)
Protection category	
M40, M40R	IP 66/68
Float damping during gas measurement recommended	
DN15...25/½"...1"	Operating pressure <0.3 bar
DN50...100/2"...4"	Operating pressure <0.2 bar

Installation conditions acc. to VDI/VDE 3513 Sheet 3

Inlet section	≥ 5 x DN
Outlet section	≥ 3 x DN

Materials

Device	Flange	Measuring tube	Float	Float guide
H250/RR	SS316L	SS316L	SS316L	SS316L
H250/PTFE	SS316 with PTFE Flare	SS316 with PTFE	PTFE	PTFE

M40	Die Cast Aluminium
M40R	Stainless steel without coating

Temperatures

For devices to be used in hazardous areas, special temperature ranges apply. These can be found in the separate instructions

Temperatures H250/M40 - mechanical indicator without power supply

	Product temperature	Ambient temperature
	[°C]	[°C]
H250/RR	-80...+300	-40...+120
H250/PTFE	-80...+70	-40...+70